

Commerce XI Maths App

Theory (Pages 37–43)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 3 - Skewness | Total questions: 55

Question 1 - MCQ · PG-37 · TH-Q1

Skewness indicates whether a distribution is

Options

- A. symmetric or asymmetric
- B. continuous or discrete only
- C. large or small only
- D. grouped or ungrouped only

Correct answer: symmetric or asymmetric

Explanation and steps

Skewness measures symmetry and, when asymmetric, its direction and extent.

Question 2 - MCQ · PG-37 · TH-Q2

A symmetric distribution has

Options

- A. zero skewness
- B. positive skewness
- C. negative skewness
- D. unit skewness

Correct answer: zero skewness

Explanation and steps

Equal balance on both sides gives zero skewness.

Question 3 - MCQ · PG-37 · TH-Q3

In a symmetric distribution, values equidistant from the mean have

Options

- A. equal frequencies
- B. zero frequencies
- C. unequal frequencies
- D. negative frequencies

Correct answer: equal frequencies

Explanation and steps

Mirrored values have equal frequencies.

Question 4 - MCQ · PG-37 · TH-Q4

X (variable)	14	16	18	20	22	24	26
f (frequency)	2	5	10	14	10	5	2

The mean of this distribution is

Options

A. 20

B. 18

C. 22

D. 24

Correct answer: 20

Explanation and steps

The frequencies balance around $x = 20$.

Question 5 - MCQ · PG-37 · TH-Q5

X (variable)	14	16	18	20	22	24	26
f (frequency)	2	5	10	14	10	5	2

The frequencies corresponding to $x = 18$ and $x = 22$ are

Options

A. 10 and 10

B. 5 and 5

C. 14 and 14

D. 2 and 2

Correct answer: 10 and 10

Explanation and steps

Both values are two units from the mean and have frequency 10.

Question 6 - MCQ · PG-37 · TH-Q6

For a symmetric distribution, the correct relation is

Options

- A. mean = median = mode
- B. mean > median > mode
- C. mode > median > mean
- D. mean = mode > median

Correct answer: mean = median = mode

Explanation and steps

All three central values coincide in a symmetric distribution.

Question 7 - MCQ · PG-37 · TH-Q7

Which measurements are usually approximately symmetric?

Options

- A. lengths of rods and weights of individuals
- B. salaries only
- C. lifespans only
- D. profits only

Correct answer: lengths of rods and weights of individuals

Explanation and steps

The textbook gives lengths and weights as typical approximate examples.

Question 8 - MCQ · PG-37 · TH-Q8

A distribution is asymmetric when values equidistant from the mean have

Options

- A. unequal frequencies
- B. equal frequencies
- C. zero values
- D. the same class mark

Correct answer: unequal frequencies

Explanation and steps

Unequal mirrored frequencies indicate asymmetry.

Question 9 - MCQ · PG-37 · TH-Q9

In a positively skewed distribution, the longer tail points towards the

Options

- A. positive end of the x-axis
- B. negative end of the x-axis
- C. y-axis only
- D. origin only

Correct answer: positive end of the x-axis

Explanation and steps

Positive skew has a longer right tail.

Question 10 - MCQ · PG-37 · TH-Q10

For a positively skewed distribution, the correct order is

Options

- A. mean > median > mode
- B. mode > median > mean
- C. mean = median = mode
- D. median > mean > mode

Correct answer: mean > median > mode

Explanation and steps

High values pull the mean furthest to the right.

Question 11 - MCQ · PG-38 · TH-Q11

Employee salaries are often

Options

- A. positively skewed
- B. negatively skewed
- C. always symmetric
- D. without dispersion

Correct answer: positively skewed

Explanation and steps

A few very high salaries create a long right tail.

Question 12 - MCQ · PG-38 · TH-Q12

X (variable)	14	16	18	20	22	24	26
f (frequency)	2	8	17	10	7	4	2

The modal value of x is

Options

- A. 18
- B. 16
- C. 20
- D. 22

Correct answer: 18

Explanation and steps

The largest frequency, 17, occurs at $x = 18$.

Question 13 - MCQ · PG-38 · TH-Q13

In a negatively skewed distribution, the longer tail is on the

Options

- A. left side
- B. right side
- C. upper side
- D. both sides equally

Correct answer: left side

Explanation and steps

The tail points towards the negative end of the x-axis.

Question 14 - MCQ · PG-38 · TH-Q14

For a negatively skewed distribution, the correct order is

Options

- A. mode > median > mean
- B. mean > median > mode
- C. mean = median = mode
- D. median > mode > mean

Correct answer: mode > median > mean

Explanation and steps

Low values pull the mean furthest to the left.

Question 15 - MCQ · PG-38 · TH-Q15

Lifespans of individuals are often

Options

- A. negatively skewed
- B. positively skewed
- C. always symmetric
- D. uniform

Correct answer: negatively skewed

Explanation and steps

The textbook gives lifespan as an example of negative skewness.

Question 16 - MCQ · PG-38 · TH-Q16

Karl Pearson's coefficient of skewness is

Options

- A. $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$
- B. $\frac{\text{Mode} - \text{Mean}}{\text{S.D.}}$
- C. $\frac{\text{Mean} - \text{Median}}{\text{Variance}}$
- D. $\frac{Q_3 - Q_1}{2}$

Correct answer: $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$

Explanation and steps

Pearson standardizes Mean – Mode by the standard deviation.

Question 17 - MCQ · PG-38 · TH-Q17

Dividing Mean – Mode by standard deviation makes the coefficient

Options

- A. independent of units
- B. equal to the variance
- C. always positive
- D. equal to the mode

Correct answer: independent of units

Explanation and steps

The standardized coefficient permits comparisons across data measured in different units.

Question 18 - MCQ · PG-38 · TH-Q18

For a positively skewed distribution, Sk_p is

Options

- A. greater than 0
- B. less than 0
- C. equal to 0
- D. always greater than 3

Correct answer: greater than 0

Explanation and steps

Mean exceeds mode, so the numerator is positive.

Question 19 - MCQ · PG-38 · TH-Q19

For a negatively skewed distribution, Sk_p is

Options

- A. less than 0
- B. greater than 0
- C. equal to 0
- D. always less than -3

Correct answer: less than 0

Explanation and steps

Mean is below mode, so the numerator is negative.

Question 20 - MCQ · PG-38 · TH-Q20

Most Pearsonian coefficients of skewness lie between

Options

- A. -1 and 1
- B. 0 and 1 only
- C. 1 and 3
- D. -10 and 10

Correct answer: -1 and 1

Explanation and steps

Most values fall between -1 and 1; all lie between -3 and 3.

Question 21 - MCQ · PG-38 · TH-Q21

All Pearsonian coefficients of skewness lie between

Options

- A. -3 and 3
- B. -1 and 1
- C. 0 and 3
- D. -10 and 10

Correct answer: -3 and 3

Explanation and steps

The textbook gives the full range as -3 to 3.

Question 22 - MCQ · PG-38 · TH-Q22

When mode is indeterminate, the empirical relation used is

Options

- A. $\text{Mean} - \text{Mode} = 3 \times (\text{Mean} - \text{Median})$
- B. $\text{Mean} - \text{Mode} = \text{Mean} - \text{Median}$
- C. $\text{Mode} - \text{Mean} = 3 \times \text{Median}$
- D. $\text{Mean} + \text{Mode} = 3 \times \text{Median}$

Correct answer: $\text{Mean} - \text{Mode} = 3 \times (\text{Mean} - \text{Median})$

Explanation and steps

The empirical relation replaces the unavailable mode.

Question 23 - MCQ · PG-39 · TH-Q23

When mode is indeterminate, Pearson's coefficient becomes

Options

- A. $3 \left(\frac{\text{Mean} - \text{Median}}{\text{S.D.}} \right)$
- B. $\frac{\text{Mean} - \text{Median}}{\text{S.D.}}$
- C. $3 \left(\frac{\text{Mode} - \text{Median}}{\text{S.D.}} \right)$
- D. $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Correct answer: $3 \left(\frac{\text{Mean} - \text{Median}}{\text{S.D.}} \right)$

Explanation and steps

Substitute the empirical relation into Pearson's formula.

Question 24 - MCQ · PG-39 · TH-Q24

Profit (₹ lakh)	10–20	20–30	30–40	40–50	50–60
No. of firms	12	18	25	10	7

The modal class is

Options

A. 30–40

B. 20–30

C. 40–50

D. 50–60

Correct answer: 30–40

Explanation and steps

The highest frequency 25 occurs in 30–40.

Question 25 - MCQ · PG-39 · TH-Q25

Profit (₹ lakh)	10–20	20–30	30–40	40–50	50–60
No. of firms	12	18	25	10	7

The mode is approximately

Options

A. 33.18

B. 32.5

C. 35

D. 11.7554

Correct answer: 33.18

Explanation and steps

Using the grouped mode formula gives 33.18.

Question 26 - MCQ · PG-39 · TH-Q26

Profit (₹ lakh)	10–20	20–30	30–40	40–50	50–60
No. of firms	12	18	25	10	7

The mean is

Options

- A. 32.5
- B. 33.18
- C. 35
- D. 30

Correct answer: 32.5

Explanation and steps

With $A = 35$ and $h = 10$, the coded mean gives 32.5.

Question 27 - MCQ · PG-39 · TH-Q27

Profit (₹ lakh)	10–20	20–30	30–40	40–50	50–60
No. of firms	12	18	25	10	7

The standard deviation is approximately

Options

- A. 11.7554
- B. 1.3819
- C. 32.5
- D. 33.18

Correct answer: 11.7554

Explanation and steps

Variance is 138.19, so $S.D. = \sqrt{138.19} = 11.7554$.

Question 28 - MCQ · PG-39 · TH-Q28

Profit (₹ lakh)	10–20	20–30	30–40	40–50	50–60
No. of firms	12	18	25	10	7

The distribution is

Options

- A. negatively skewed
- B. positively skewed
- C. symmetric
- D. without skewness measure

Correct answer: negatively skewed

Explanation and steps

$$S_{kp} = \frac{32.5 - 33.18}{11.7554} = -0.0578.$$

Question 29 - MCQ · PG-39 · TH-Q29

Bowley's coefficient of skewness is based on

Options

- A. quartiles
- B. mean and mode only
- C. variance only
- D. range only

Correct answer: quartiles

Explanation and steps

Bowley's measure uses Q1, Q2 and Q3.

Question 30 - MCQ · PG-39 · TH-Q30

Bowley's coefficient is

Options

- A. $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$
- B. $\frac{Q_3 - Q_1}{2}$
- C. $\frac{Q_3 + Q_1}{2Q_2}$
- D. $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$

Correct answer: $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Explanation and steps

This is the standard Bowley formula.

Question 31 - MCQ · PG-39 · TH-Q31

For positive skewness, which comparison is correct?

Options

- A. $Q_3 - Q_2 > Q_2 - Q_1$
- B. $Q_3 - Q_2 < Q_2 - Q_1$
- C. $Q_3 - Q_2 = Q_2 - Q_1$
- D. $Q_3 = Q_2 = Q_1$

Correct answer: $Q_3 - Q_2 > Q_2 - Q_1$

Explanation and steps

The upper half is more spread out in positive skewness.

Question 32 - MCQ · PG-40 · TH-Q32

For negative skewness, which comparison is correct?

Options

- A. $Q_3 - Q_2 < Q_2 - Q_1$
- B. $Q_3 - Q_2 > Q_2 - Q_1$
- C. $Q_3 - Q_2 = Q_2 - Q_1$
- D. $Q_3 = Q_1$

Correct answer: $Q_3 - Q_2 < Q_2 - Q_1$

Explanation and steps

The lower half is more spread out in negative skewness.

Question 33 - MCQ · PG-40 · TH-Q33

Bowley's coefficient for a symmetric distribution is

Options

- A. 0
- B. 1
- C. -1
- D. 3

Correct answer: 0

Explanation and steps

Equal quartile distances make the numerator zero.

Question 34 - MCQ · PG-40 · TH-Q34

Bowley's coefficient lies between

Options

- A. -1 and 1
- B. -3 and 3
- C. 0 and 1 only
- D. -10 and 10

Correct answer: -1 and 1

Explanation and steps

Bowley's coefficient is bounded by -1 and 1.

Question 35 - MCQ · PG-40 · TH-Q35

Bowley's coefficient can be computed for distributions with

Options

- A. open-ended extreme classes
- B. no quartiles
- C. negative frequencies
- D. no median

Correct answer: open-ended extreme classes

Explanation and steps

Extreme open classes do not affect the quartiles when quartile classes are internal.

Question 36 - MCQ · PG-40 · TH-Q36

If $Q_1 = 80$, $Q_2 = 100$ and $Q_3 = 120$, Bowley's coefficient is

Options

- A. 0
- B. 0.5
- C. -0.5
- D. 1

Correct answer: 0

Explanation and steps

The numerator is $120 + 80 - 200 = 0$.

Question 37 - MCQ · PG-40 · TH-Q37

Wages	Below 300	300–400	400–500	500–600	600–700	Above 700
No. of workers	5	8	18	35	27	7

The total number of workers is

Options

- A. 100
- B. 93
- C. 105
- D. 700

Correct answer: 100

Explanation and steps

The six frequencies sum to 100.

Question 38 - MCQ · PG-40 · TH-Q38

Wages	Below 300	300–400	400–500	500–600	600–700	Above 700
No. of workers	5	8	18	35	27	7

Q₁ lies in the class

Options

A. 400–500

B. 300–400

C. 500–600

D. 600–700

Correct answer: 400–500

Explanation and steps

The 25th observation lies after cumulative frequency 13 and before 31.

Question 39 - MCQ · PG-40 · TH-Q39

Wages	Below 300	300–400	400–500	500–600	600–700	Above 700
No. of workers	5	8	18	35	27	7

Q₂ lies in the class

Options

A. 500–600

B. 400–500

C. 600–700

D. 300–400

Correct answer: 500–600

Explanation and steps

The 50th observation lies after cumulative frequency 31 and before 66.

Question 40 - MCQ · PG-40 · TH-Q40

Wages	Below 300	300–400	400–500	500–600	600–700	Above 700
No. of workers	5	8	18	35	27	7

Q₃ lies in the class

Options

- A. 600–700
- B. 500–600
- C. 400–500
- D. Above 700

Correct answer: 600–700

Explanation and steps

The 75th observation lies after cumulative frequency 66 and before 93.

Question 41 - MCQ · PG-40 · TH-Q41

Wages	Below 300	300–400	400–500	500–600	600–700	Above 700
No. of workers	5	8	18	35	27	7

Bowley's coefficient is approximately

Options

- A. –0.05
- B. 0.05
- C. –0.5
- D. 0

Correct answer: –0.05

Explanation and steps

Using $Q_1 = 466.67$, $Q_2 = 554.28$ and $Q_3 = 633.33$ gives about –0.05.

Question 42 - MCQ · PG-41 · TH-Q42

If mean = 100, median = 98.5 and S.D. = 9, the mode is

Options

- A. 95.5
- B. 98.5
- C. 100
- D. 104.5

Correct answer: 95.5

Explanation and steps

Mode = Mean - $3 \times (\text{Mean} - \text{Median}) = 95.5$.

Question 43 - MCQ · PG-41 · TH-Q43

If mean = 100, median = 98.5 and S.D. = 9, Sk_p is

Options

- A. 0.5
- B. -0.5
- C. 1.5
- D. 0

Correct answer: 0.5

Explanation and steps

$$Sk_p = 3 \left(\frac{100 - 98.5}{9} \right) = 0.5.$$

Question 44 - MCQ · PG-41 · TH-Q44

If mode is greater than mean by 7 and variance is 100, the distribution is

Options

- A. negatively skewed
- B. positively skewed
- C. symmetric
- D. indeterminate

Correct answer: negatively skewed

Explanation and steps

S.D. = 10 and Mean - Mode = -7, giving $Sk_p = -0.7$.

Question 45 - MCQ · PG-42 · TH-Q45

If mean = 214, mode = 218 and variance = 196, Sk_p is approximately

Options

- A. -0.2857
- B. 0.2857
- C. -4
- D. 14

Correct answer: -0.2857

Explanation and steps

$$S.D. = 14 \text{ and } Sk_p = \frac{214 - 218}{14}.$$

Question 46 - MCQ · PG-42 · TH-Q46

If $Q_1 = 35$, $Q_2 = 40$ and the distribution is symmetric, Q_3 is

Options

- A. 45
- B. 40
- C. 50
- D. 75

Correct answer: 45

Explanation and steps

Symmetry gives $Q_3 - Q_2 = Q_2 - Q_1 = 5$.

Question 47 - MCQ · PG-42 · TH-Q47

If $Q_1 = 15$, $Q_2 = 21$ and $Q_3 = 29$, Bowley's coefficient is approximately

Options

- A. 0.143
- B. -0.143
- C. 0.5
- D. 1

Correct answer: 0.143

Explanation and steps

$$Sk_b = \frac{29 + 15 - 42}{29 - 15} = \frac{2}{14}.$$

Question 48 - MCQ · PG-42 · TH-Q48

For a negatively skewed distribution, the required relation is

Options

- A. Mean < Mode
- B. Mean > Mode
- C. Mean = Mode
- D. Mean = 0

Correct answer: Mean < Mode

Explanation and steps

Since S.D. is positive and $Skp < 0$, Mean – Mode < 0.

Question 49 - MCQ · PG-42 · TH-Q49

If a distribution is symmetric, its quartiles satisfy

Options

- A. $2Q_2 = Q_1 + Q_3$
- B. $Q_2 = Q_1 + Q_3$
- C. $Q_3 = Q_2 + Q_1$
- D. $Q_1 = Q_2 = Q_3$ only

Correct answer: $2Q_2 = Q_1 + Q_3$

Explanation and steps

Equal quartile distances give $Q_3 - Q_2 = Q_2 - Q_1$.

Question 50 - MCQ · PG-43 · TH-Q50

If Q_1 , Q_2 and Q_3 are in arithmetic progression, the distribution is

Options

- A. symmetric
- B. positively skewed
- C. negatively skewed
- D. always bimodal

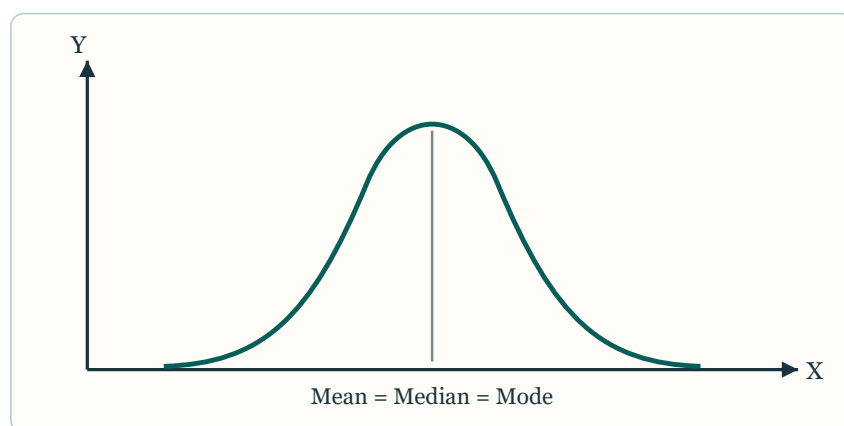
Correct answer: symmetric

Explanation and steps

An A.P. gives $Q_3 - Q_2 = Q_2 - Q_1$, hence $Skb = 0$.

Question 51 - MCQ · PG-37 · TH-Q51

Observe the distribution curve shown below. Which relationship is represented?



Options

- A. Mean = Median = Mode
- B. Mean > Median > Mode
- C. Mode > Median > Mean
- D. Mean = Mode > Median

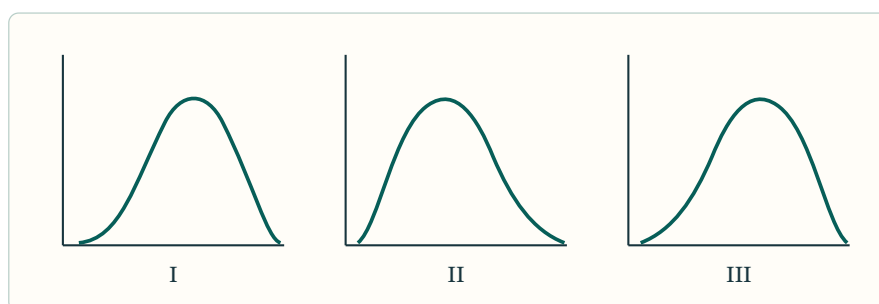
Correct answer: Mean = Median = Mode

Explanation and steps

The curve is symmetric, so mean, median and mode coincide.

Question 52 - MCQ · PG-38 · TH-Q52

Identify the distributions represented by curves I, II and III respectively.



Options

- A. Symmetric, positively skewed, negatively skewed
- B. Positively skewed, symmetric, negatively skewed
- C. Negatively skewed, positively skewed, symmetric
- D. Symmetric, negatively skewed, positively skewed

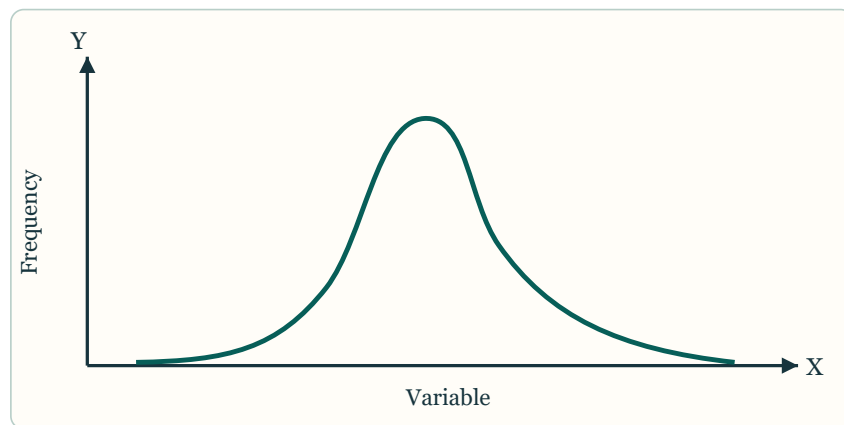
Correct answer: Symmetric, positively skewed, negatively skewed

Explanation and steps

Curve I is balanced, curve II has a longer right tail, and curve III has a longer left tail.

Question 53 - MCQ · PG-38 · TH-Q53

Observe the frequency curve shown below. The distribution is



Options

- A. positively skewed
- B. negatively skewed
- C. symmetric
- D. uniform

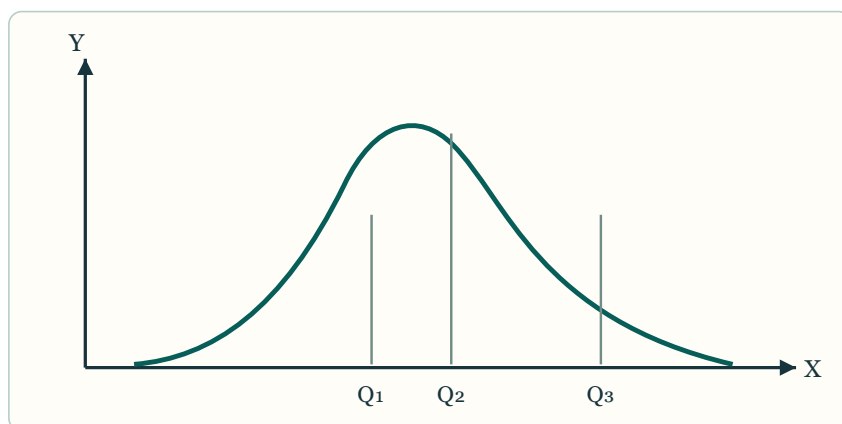
Correct answer: positively skewed

Explanation and steps

The longer tail extends towards the higher values on the right.

Question 54 - MCQ · PG-40 · TH-Q54

For the distribution shown below, which relationship is correct?



Options

- A. $Q_3 - Q_2 > Q_2 - Q_1$
- B. $Q_3 - Q_2 < Q_2 - Q_1$
- C. $Q_3 - Q_2 = Q_2 - Q_1$
- D. $Q_1 = Q_2 = Q_3$

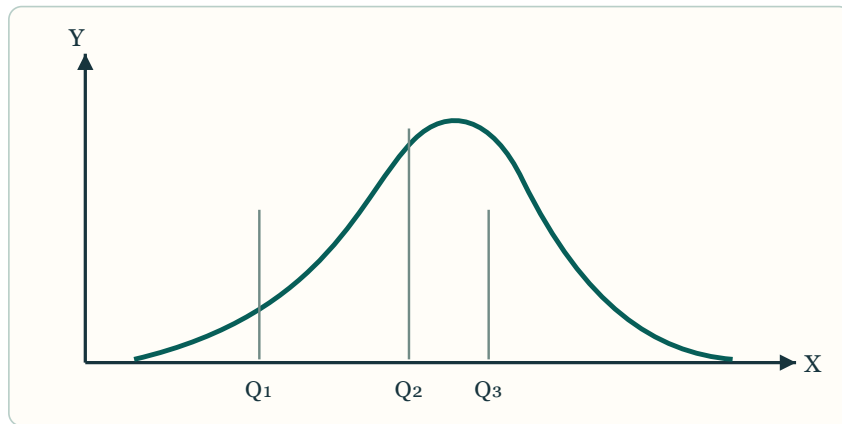
Correct answer: $Q_3 - Q_2 > Q_2 - Q_1$

Explanation and steps

In positive skewness, the distance from Q_2 to Q_3 is greater than the distance from Q_1 to Q_2 .

Question 55 - MCQ · PG-40 · TH-Q55

For the distribution shown below, which relationship is correct?



Options

- A. $Q_3 - Q_2 < Q_2 - Q_1$
- B. $Q_3 - Q_2 > Q_2 - Q_1$
- C. $Q_3 - Q_2 = Q_2 - Q_1$
- D. $Q_1 = Q_2 = Q_3$

Correct answer: $Q_3 - Q_2 < Q_2 - Q_1$

Explanation and steps

In negative skewness, the distance from Q_1 to Q_2 is greater than the distance from Q_2 to Q_3 .